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Can a Buzz Answer That? How Much of What Blind People Ask a Visual Assistant Fits a Vibration Vocabulary

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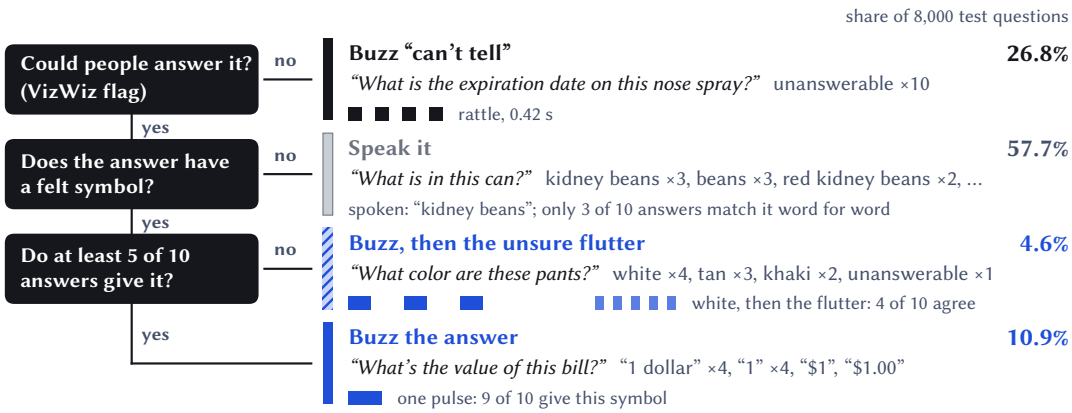


Fig. 1. How each reply is routed. Percentages are shares of the 8,000 held-out test questions. Each route shows a real test question, its crowd answers and the rhythm a phone would play (bars show vibration on, all drawn to one time scale).

Visual assistants answer blind people by speech, occupying the hearing they rely on. Could a vibration rhythm answer instead? For 32,842 VizWiz questions, we routed the crowd's answer to a felt symbol (can't tell, yes, no, number, colour) or speech, fixing all rules before opening 8,000 held-out questions. Rhythms could encode an informative answer for only 15.5% of these (10.9% with at least half agreeing, 4.6% marked unsure) and "can't tell" for 26.8% more. Rhythms were not reliably shorter than estimated speech. An off-the-shelf visual question answering model not trained to abstain confidently buzzed an answer for 20% of photos the crowd marked unanswerable. A felt channel should carry failure and uncertainty first, behind an assistant that can say it cannot tell. Recognition is untested; an open Android prototype is built to play rhythms and run identification tests.

CCS Concepts: • **Human-centered computing** → **Accessibility technologies**; *Haptic devices.*

Additional Key Words and Phrases: blind and low vision, visual question answering, vibrotactile feedback, tactons, uncertainty, VizWiz

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1 Introduction

When blind and low-vision (BLV) people ask what is in a photo they took, a visual assistant answers by voice, whether it is a crowd worker [8, 9] or an AI model [17]. Speech competes with the ambient sounds BLV people use to travel safely [16], and people nearby can hear it [30]. Yet many answers are very short: a colour, a yes, the value of a banknote.

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Such short messages suit vibration patterns, or tactons [11, 12, 29]. Tactile aids for BLV people mostly guide movement or render braille [28, 31, 32], and we found no study of how many real answers a small vibration vocabulary could cover.

Answers are also often uncertain. Crowd workers disagree [7, 19], many photos cannot be answered at all [15, 20], and blind users miss many AI errors and have to find their own ways to check answers [1, 21, 27]. So a vibration channel would need to say "I can't tell" and "I'm not sure" too.

We test how far such a channel could go with 32,842 questions that blind people asked about their own photos, each with 10 crowd answers [20]. We contribute (1) an estimate of which replies fit a rhythm vocabulary of can't tell, yes, no, numbers and colours (Fig. 1), and of how consistently people give them; (2) a check with a real visual question answering (VQA) model, showing why the channel needs a failure signal the model can produce; and (3) an open prototype, Felt Answers, for testing the rhythms with people.

2 Related Work

What blind people ask. Most VizWiz questions ask to identify an object, describe it (often its colour) or read text [8, 9]. Yes/no and number answers are rare, and a quarter to a third of the questions cannot be answered from the photo [20]. Later work labelled why answers differ [7], what is wrong with the photos [15] and which vision skills a question needs [35]. BLV people now use multimodal AI assistants every day and run into made-up answers [17, 22, 34].

Touch. Tactons encode messages in rhythm, the parameter people identify most reliably, and in roughness [11, 12]. Counted pulses can convey digits on the wrist [29], and smartwatches already tap out the time [4]. People identify a handful of vibrotactile cues without training, and more with practice [5, 14, 26]. Vibration has rendered braille [23, 28] and colour [13], while camera aids guide aiming mostly with speech or tones [24, 32].

3 Method

Data. We used all VizWiz-VQA questions: training (20,523), validation (4,319) and test (8,000), whose 328,420 crowd answers are now public. For training and validation we also used labels for why answers differ [7] and for image-quality flaws [15], counting a label if at least 2 of 5 annotators chose it. Answers were normalised with the official VQA rules [3].

The assistant's answer. We wanted to study how an answer is delivered, not whether it is right, so the assistant says what the crowd says: "can't tell" if VizWiz marks the question unanswerable, and otherwise the most frequent answer. This bounds how correct a vibration reply could be, but not how many questions could get one.

Vocabulary. All rules were set on the training and validation splits before the test split was opened (Table 1). Tier 0 is meant to need little learning: can't tell, yes, no, and numbers up to 999, played digit by digit as counted pulses. Tier 1 adds the 11 basic colour terms [6], one or two per answer and without modifiers ("light blue" is spoken), as rhythms of 3 or 4 elements, shorter for frequent colours. Tier 2 adds K learned symbols for other frequent answers. The categories set coverage; the rhythms set only duration and which codes may be confused. Pulses of 100–600 ms with gaps of at least 150 ms suit phone actuators [2] and counting [29]. Two signals break this rule on purpose: the can't-tell rattle and the unsure flutter are fast pulse trains meant to feel rough, not countable [12].

Routing. An answer without a symbol is spoken. An answer with one (a felt answer) is played alone if at least half of the 10 crowd answers map to that symbol (agreement $k \geq 5$), and is otherwise followed by the flutter. Can't tell is never marked unsure. We estimated speech time from word counts at 180 words per minute, the top of typical speaking rates, and at 334, the mean maximum intelligible rate for visually impaired listeners [10]. Both rates favour speech. Neither estimate includes latency: decoding and motor time for rhythms, synthesis for speech. In speech, unsure answers start with "I'm not sure, but", and can't tell is "I can't tell, please take another photo" or just "Can't tell".

Table 1. The vibration vocabulary. s: 100 ms pulse; l: 300 ms pulse; elements 150 ms apart; symbols 500 ms apart. Examples are drawn to one time scale (bars show vibration on): the number is 3, the colour is green, and unsure shows the flutter alone.

Meaning	Rhythm	Example	Duration
Can't tell	rattle: four 60 ms pulses, 60 ms apart		0.42 s
Yes; no	l; s s		0.30; 0.35 s
Number	digits as counted 150 ms pulses (200 ms gaps, 450 after five), 600 ms between digits; 0: one 600 ms pulse		0.15–10.8 s
Colour	white sss, grey ssl, black sls, blue lss, red sll, brown lsl, pink lls, green lll, purple ssss, yellow sssl, orange ssls		0.60–2.75 s
Unsure	flutter after the answer: five 40 ms pulses, 40 ms apart		+0.86 s

A real model. Real assistants can be wrong, so we ran ViLT-B/32 fine-tuned on VQAv2 [18, 25] zero-shot on all 8,000 test questions. ViLT picks one of 3,129 answers, none of which is "unanswerable". Its confidence p is the sigmoid of the top logit, as it was trained on soft VQA scores; with the same symbol rules, a confident buzz ($p \geq 0.5$) is played alone and others with the flutter. A buzz on an answerable photo counts as right if at least 3 of the 10 crowd answers map to its symbol. The analysis plan was fixed before inference, with one later addendum. ViLT is older and weaker than current assistants and trained neither on VizWiz nor to abstain (VQA accuracy 17% here), but it is open and runs offline.

Prototype. Felt Answers is an offline web page for Chrome on Android with four modes. *Ask* plays 228 real test questions as rhythm or speech, *Learn* lists every code, *Test* runs identification trials in shuffled blocks with a confusion matrix, and *Check* lets a tester rate each pattern on a table and in the hand and logs vibration timing. Results export as CSV or JSON, and nothing is sent online. Its JavaScript routing matched the Python analysis on all 32,842 questions.

4 Results

A buzz could cover 42.3% of replies, but only 15.5% would be informative answers. Of the 8,000 test questions, 26.8% were marked unanswerable and would get can't tell, and 15.5% got a felt answer (colour 8.6%, no 2.6%, yes 2.4%, number 1.9%), 10.9% without the flutter and 4.6% with it (Fig. 1). Without colours, felt answers fell to 6.8%. Learned symbols helped slowly against a long tail of 6,397 distinct training answers: even 128 symbols, far more than people identify without training [5, 14], raised felt replies only to 55.5%, although felt answers nearly doubled, to 28.6% (Fig. 2a).

Closed answers agreed more, partly by construction. Among 17,925 answerable training and validation questions, at least half of the answerers gave the same symbol for 79% of yes, 75% of no, 69% of number and 66% of colour answers. For spoken answers, at least half gave the same string only 42% of the time (Fig. 2b). The comparison favours symbols, since open answers can differ in wording without differing in meaning [7]. On the test set, 29% of felt answers fell below $k = 5$ and carried the flutter. Even can't tell was contested: for 12% of can't-tell questions (261 of 2,147), only 4 or 5 of the 10 answerers said the question was unanswerable.

A model not trained to abstain buzzed on many unanswerable photos. ViLT buzzed on 35.3% of test questions, 23.3% confidently. On the 2,147 photos the crowd marked unanswerable it buzzed 38% of the time, 20% confidently, and gave an abstention-like answer such as "unknown" for only 5% of them (unplanned; such answers are spoken). Its confidence barely separated unanswerable from answerable photos (AUROC 0.57, also unplanned), but it did carry some signal: on answerable photos, confident buzzes were right 41% of the time and flutter-marked ones 6%. Of the 1,867 confident buzzes, 68% were wrong: fewer than 3 of 10 answerers shared the symbol, or the photo was unanswerable (Fig. 2c). Thresholds of 0.3 or 0.7, softmax confidence or a looser definition of right gave 64–72%.

Rhythms were not reliably shorter than speech. At 180 words per minute the rhythm was shorter than speech for 75% of felt replies, mostly because of can't tell: the rattle lasts 0.42 s, against 0.67 s for just the words "can't tell". Yes was only slightly shorter (0.30 s against 0.33 s), and no slightly longer (0.35 s). Against "can't tell", the share fell to 11% if

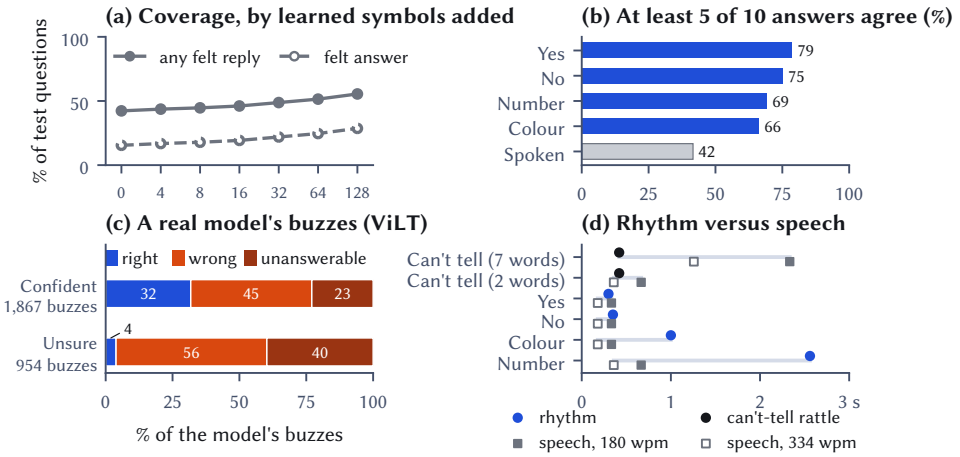


Fig. 2. (a) Test questions with any felt reply (including can't tell) or a felt answer, as learned symbols are added. (b) Answerable training and validation questions where at least 5 of 10 answerers agree (on the symbol, or for spoken answers the exact string). (c) ViLT's test buzzes: right, wrong, or on an unanswerable photo. (d) Median rhythm duration against estimated speech time.

the rattle kept our 100 ms pulse rule (0.85 s), and to 0.4% at 334 words per minute (0.36 s). Colour (1.00 s) and number (2.56 s) rhythms took longer to play than to say (Fig. 2d). The stronger case for touch is the free ear: counting only can't tell and confident felt answers, touch would have kept hearing free for 37.7% of questions.

5 Design Implications

Use touch for failure, doubt and short closed answers. The most frequent felt message is a failure, and without the flutter, touch would overstate confidence, an error blind users find hard to catch [21]. Yes/no answers, numbers and colours are candidates for touch, since answerers agree on them more often and blind users prefer short answers to questions about one attribute [22]. **Keep reading, naming and description in speech.** Most replies (57.7%) had no symbol. **The assistant must be able to abstain.** ViLT rarely did, so trained abstention or photo-quality detection has to sit behind a felt "can't tell" [15, 33]. A retake rhythm could not reliably say why: blur, framing or darkness was flagged for 87% of unanswerable and 67% of answerable labelled photos. **Some codes differ only in timing:** no and two by 50 ms in pulse and gap, yes and one in pulse length. Keyword rules on the question's wording predicted the category of 83% of felt answers, so context may tell them apart; if not, each category could open with its own element.

Limitations. Our model is a weak zero-shot probe. No one has felt these rhythms in a study yet: rendering on phones, identification, learning and preference over speech are untested. VizWiz questions are over a decade old and from one community, speech times are estimates, and Web Vibration works only on Android.

Conclusion. Over a quarter of VizWiz questions could get a felt "can't tell" but few an informative answer, so a vibration channel should start with failure, behind a model that can abstain.

Ethics and use of AI. We used public, de-identified data (CC BY 4.0); there were no participants. An AI assistant (Claude, Anthropic) helped search the literature, wrote and ran the analysis, figure and prototype code under the author's direction, and drafted the text; the author is responsible for the content.

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